Chapter 18

How to use JavaScript to enhance your web pages

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Objectives

Applied

1. Given an HTML document, use JavaScript to enhance the page so it includes the current date, current year, or a Print button.
2. Given an HTML document and tested JavaScript code, use the code to enhance a web page so it includes an image swap or slide show.
3. Use the Internet to find tested JavaScript and jQuery code that you can use to enhance a web page.

Knowledge

1. Describe how JavaScript works.
2. Describe the Document Object Model (DOM) and explain what happens when DOM scripting is used to change the DOM.
3. In general terms, describe how you script the DOM and how the $ sign is commonly used for DOM scripting.

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Objectives (continued)

4. Describe the way that JavaScript event handlers work with DOM scripting.
5. Describe jQuery and the two ways you can include it in a web page.
6. Describe the syntax for calling jQuery methods, including the use of the $ sign, selectors, and the dot operator.
7. Describe the syntax for coding a jQuery event handler, including the use of an event method and function.
8. Describe the use of the jQuery ready method.
9. In general terms, describe how you use jQuery for functions like image swaps and slide shows.

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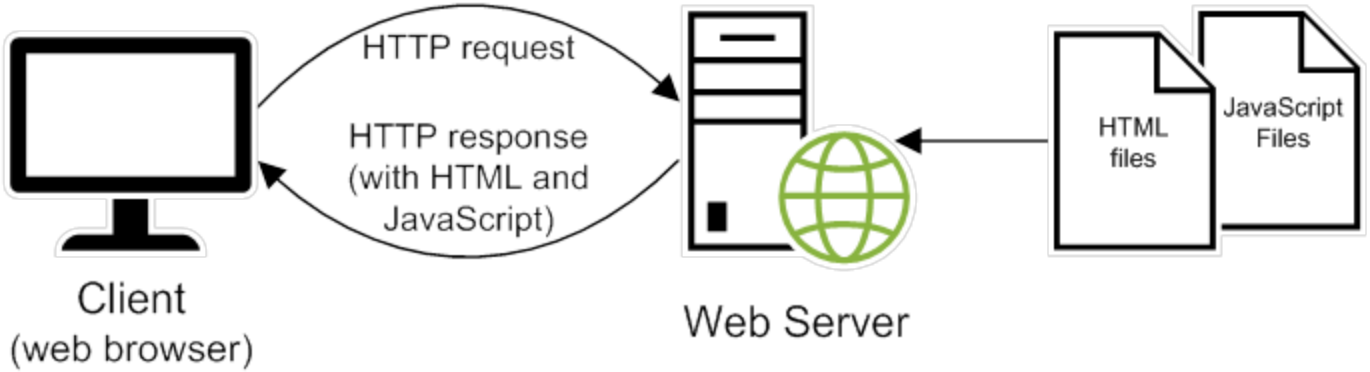
How JavaScript fits into the client/server architecture



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JavaScript that gets the current date and year

```
<p>
    <script>
        var today = new Date();
        document.write("Current date: ");
        document.write(today.toDateString());
    </script>
</p>
<p>
    <script>
        var today = new Date();
        document.write("&copy; ");
        document.write(today.getFullYear());
        document.write(", San Joaquin Valley Town Hall")
    </script>
</p>
```


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The JavaScript for current date and year in a web browser

Current date: Wed Sep 08 2021

© 2021, San Joaquin Valley Town Hall

A script element in the head section that loads an external JavaScript file

```
<script src="set_date.js"></script>
```


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JavaScript embedded in the head section

```
<head>
    ...
    <script>
        var $ = function (id) {
            return document.getElementById(id);
        }
        window.onload = function() {
            var today = new Date();
            $("date").firstChild.nodeValue =
                "Current date: " + today.toDateString();
        }
    </script>
</head>
```

JavaScript embedded in the body

```
<p>
    <script>
        var today = new Date();
        document.write("Current date: ");
        document.write(today.toDateString());
    </script>
</p>
```


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The code for a web page

```
<!DOCTYPE html>
<html>
<head>
    <title>Join Email List</title>
</head>
<body>
    <h1>Please join our email list</h1>
    <form id="email_form" name="email_form"
          action="join.html" method="get">
        <label for="email_address">Email Address:</label>
        <input type="text" id="email_address">
        <span id="email_error">*</span>
    </form>
</body>
</html>
```


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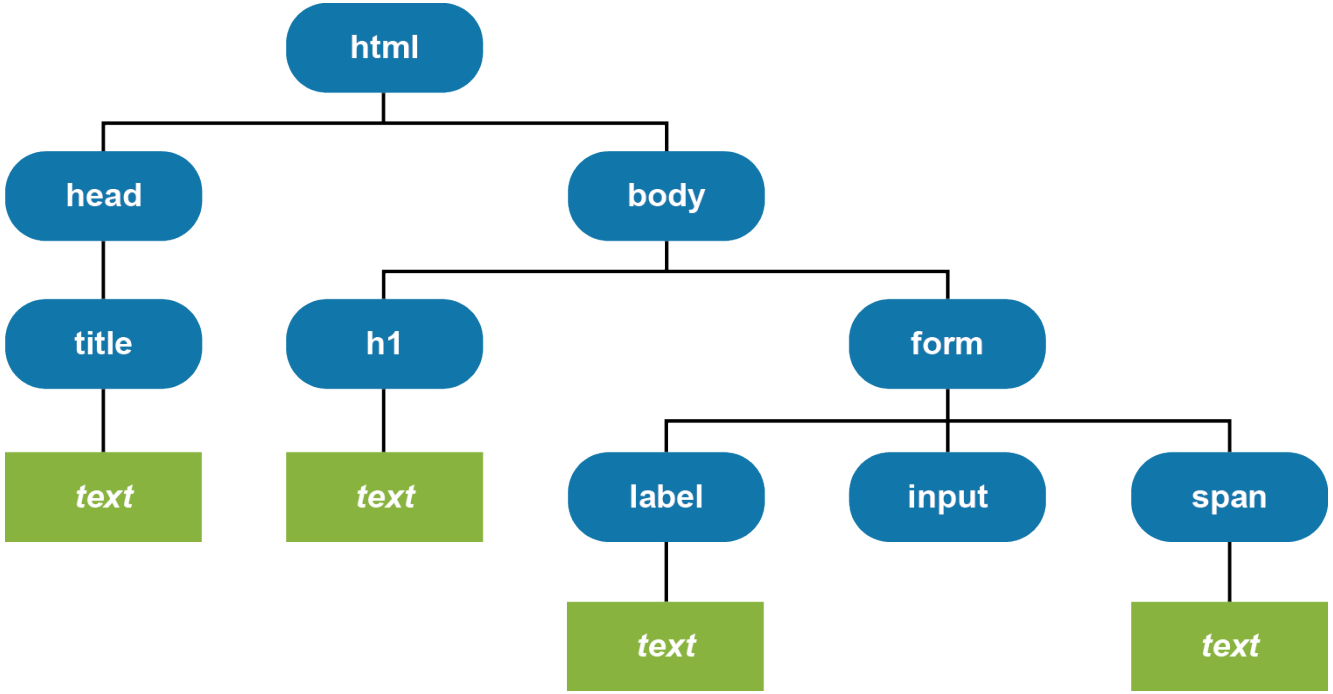
The DOM for the web page



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DOM nodes commonly used in DOM scripting

```
Element
Attr
Text
```

The DOM nodes that you commonly use in DOM scripting

Type	Description
Element	An element in the web page. It can have Element, Text, and Comment nodes as child nodes.
Attr	An attribute of an element. Although it is attached to an Element node, it isn't considered a child node. It can have a Text node as a child node.
Text	The text for an element or attribute. It can't have a child node.

The code for a web page

```
<h1>Please join our email list</h1>
<form id="email_form" name="email_form" action="join.html" method="get">
    <label for="email_address">Email Address:</label>
    <input type="text" id="email_address">
    <span id="email_error">*</span>
</form>
```


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The document object

Methods of the document object

```
getElementById(id)
write(string)
```

Examples of document methods

```
// returns the object for the HTML element
var rateBox = document.getElementById("rate");

// writes a string into the document
document.write("Today is " + today.toDateString());
```

A standard $ function that gets the object for an element by using its id

```
var $ = function (id) {
    return document.getElementById(id);
}
```


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Scripting the DOM

Three properties for scripting the DOM

```
value
firstChild
nodeValue
```

How to get the text of an HTML element

```
var emailAddress = $("email_address").value;
```

How to set the text of an HTML element

```
$("email_error").firstChild.nodeValue =
    "Entry is invalid.";
```


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A page that uses a JavaScript button for printing

Print the Page

Murach's JavaScript and jQuery

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The JavaScript in an external printPage.js file

```
var $ = function (id) {
    // this function returns the object for the element
    return document.getElementById(id);
}
var printPage = function() {
    // this is the event handler
    // for the click event of the button
    window.print();
}
window.onload = function() {
    // this is the event handler for the onload event
    $("printButton").onclick = printPage;
}
```

HTML that includes the JavaScript file

```
<script src="printPage.js"></script>
```

HTML for the Print the Page button

```
<input type="button" id="printButton"
       value="Print the Page">
```


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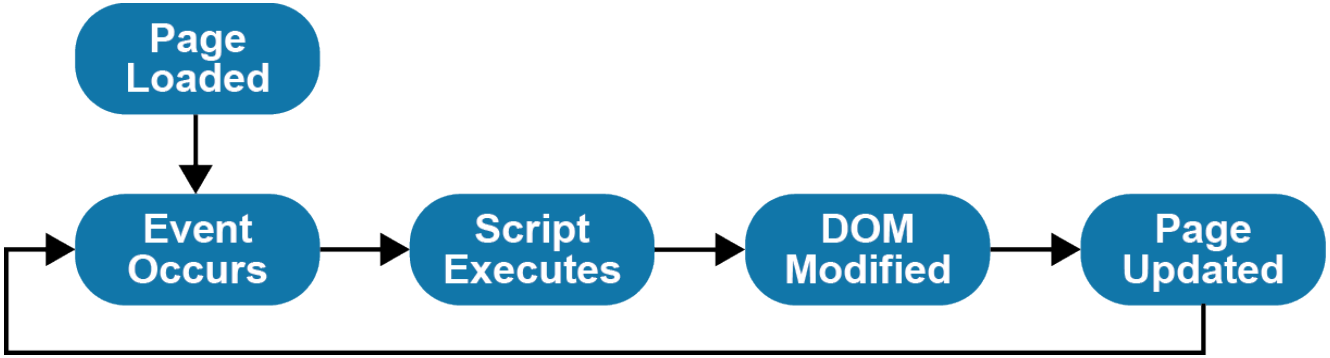
The DOM event cycle



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The Email List with JavaScript for data validation

Please join our email list

Email Address: zak@yahoo.com

Re-enter Email Address: zak@yahoo This entry must equal first entry.

First Name: This field is required.

Join our List

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The HTML file for the Email List application

```
<!DOCTYPE html>
<html>
<head>
    <meta charset="UTF-8">
    <title>Join Email List</title>
    <link rel="stylesheet" href="email_list.css">
    <script src="email_list.js"></script>
</head>
<body>
    <main>
        <h1>Please join our email list</h1>
        <form id="email_form" name="email_form"
              action="join.html" method="get">
            <label for="email_address1">
                Email Address:</label>
            <input type="text" id="email_address1"
                   name="email_address1">
            <span id="email_address1_error">*</span><br>
```


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The HTML file for the Email List app (continued)

```
            <label for="email_address2">
                Re-enter Email Address:</label>
            <input type="text" id="email_address2"
                   name="email_address2">
            <span id="email_address2_error">*</span><br>

            <label for="first_name">First Name</label>
            <input type="text" id="first_name"
                   name="first_name">
            <span id="first_name_error">*</span><br>

            <label> </label>
            <input type="button" id="join_list"
                   value="Join our List">
        </form>
    </main>
</body>
</html>
```

The code for the JavaScript file (email_list.js)

```
var $ = function (id) {
    return document.getElementById(id);
}
var joinList = function () {
    var emailAddress1 = $("email_address1").value;
    var emailAddress2 = $("email_address2").value;
    var isValid = true;

    if (emailAddress1 == "") {
        $("email_address1_error").firstChild.nodeValue =
            "This field is required.";
        isValid = false;
    }
    else {
        $("email_address1_error").firstChild.nodeValue =
            ""; }
```


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The code for the email_list.js file (continued)

```
    if (emailAddress1 !== emailAddress2) {
            $("email_address2_error").firstChild.nodeValue =
                "This entry must equal first entry.";
            isValid = false;
        }
        else {
            $("email_address2_error").firstChild.nodeValue =
                ""; }

        ...

        if (isValid) {
            // submit the form if all entries are valid
            $("email_form").submit();
        }
    }
    window.onload = function () {
        $("join_list").onclick = joinList;
    }
```


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The jQuery website at jQuery.com

jquery.com

Plugins | Contribute | Events | Support | JS Foundation

jQuery
write less, do more.

Your donations help fund the continued development and growth of **jQuery**.

SUPPORT THE PROJECT

Download | API Documentation | Blog | Plugins | Browser Support | Search

Lightweight Footprint
Only 30kB minified and gzipped. Can also be included as an AMD module

CSS3 Compliant
Supports CSS3 selectors to find elements as well as in style property manipulation

Cross-Browser
Chrome, Edge, Firefox, IE, Safari, Android, iOS, and more

Download jQuery v3.6.0
The 1.x and 2.x branches no longer receive patches.

View Source on GitHub →
How jQuery Works →

What is jQuery?

jQuery is a fast, small, and feature-rich JavaScript library. It makes things like HTML document traversal and manipulation, event handling, animation, and Ajax much simpler with an easy-to-use API that works across a multitude of browsers. With a combination of versatility and extensibility, jQuery has changed the way that millions of people write JavaScript.

Resources

- jQuery Core API Documentation
- jQuery Learning Center
- jQuery Blog
- Contribute to jQuery

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What jQuery offers

- Dozens of methods that make it easier to add JavaScript features to your web pages
- Methods that are tested for cross-browser compatibility

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How to include jQuery files in a web page

How to include the jQuery file from your computer

```
<script src="jquery-3.6.0.js"></script>
```

How to include jQuery from a CDN

```
<script src="http://code.jquery.com/jquery-3.6.0.min.js">
</script>
```


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How to code jQuery selectors

By element type

```
$("h2")
```

By id

```
$("#email_address")
```

By class attribute

```
$(".warning")
```


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How to call jQuery methods

How to get the value from a text box

```
var emailAddress = $("#email_address").val();
```

How to set the text in an element

```
$("#email_address_error").text(
    "Email address is required");
```

How to set the text for the next sibling

```
$("#email_address").next().text(
    "Email address is required");
```


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How to code jQuery event methods

How to code the ready event method

```
$(document).ready(function() {
    alert("The DOM is ready");
});
```

How to code the click event method for all h2 elements

```
$("h2").click(function() {
    alert("This heading has been clicked");
});
```

How to use the click event method within the ready event method

```
$(document).ready(function() {
    $("h2").click(function() {
        alert("This heading has been clicked");
    }); // end of click event handler
}); // end of ready event handler
```


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The Email List application using jQuery

Please join our email list

Email Address: zak@yahoo.com

Re-enter Email Address: zak@yahoo This entry must equal first entry.

First Name: This field is required.

Join our List

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The HTML for the Email List with jQuery app

```
<!DOCTYPE html>
<html>
<head>
    <meta charset="UTF-8">
    <title>Join Email List</title>
    <link rel="stylesheet" href="email_list.css">
    <script
        src="http://code.jquery.com/jquery-3.6.0.min.js">
    </script>
    <script src="email_list.js"></script>
</head>
<body>
    <main>
        <h1>Please join our email list</h1>
        <form id="email_form" name="email_form"
              action="join.html" method="get">
            <label for="email_address1">
                Email Address:</label>
            <input type="text" id="email_address1"
                   name="email_address1">
            <span>*</span><br>
```

The HTML for the Email List jQuery app (cont.)

```
<label for="email_address2">
                Re-enter Email Address:</label>
            <input type="text" id="email_address2"
                   name="email_address2">
            <span>*</span><br>

            <label for="first_name">First Name:</label>
            <input type="text" id="first_name"
                   name="first_name">
            <span>*</span><br>

            <label> </label>
            <input type="button" id="join_list"
                   value="Join our List">
        </form>
    </main>
</body>
</html>
```

The jQuery for the Email List application (part 1)

```
$(document).ready(function() {
    $("#join_list").click(function() {
        var emailAddress1 = $("#email_address1").val();
        var emailAddress2 = $("#email_address2").val();
        var isValid = true;

        // validate the first email address
        if (emailAddress1 == "") {
            $("#email_address1").next().text(
                "This field is required.");
            isValid = false;
        }
        else {
            $("#email_address1").next().text("");
        }
```


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The jQuery for the Email List application (part 2)

```
// validate the second email address
        if (emailAddress2 == "") {
            $("#email_address2").next().text(
                "This field is required.");
            isValid = false;
        }
        else if (emailAddress1 != emailAddress2) {
            $("#email_address2").next().text(
                "This entry must equal first entry.");
            isValid = false;
        }
        else {
            $("#email_address2").next().text("");
        }
```


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The jQuery for the Email List application (part 3)

```
    // validate the first name entry
            if ($("#first_name").val() == "") {
                $("#first_name").next().text(
                    "This field is required.");
                isValid = false;
            }
            else {
                $("#first_name").next().text("");
            }

            // submit the form if all entries are valid
            if (isValid) {
                $("#email_form").submit();
            }
        }); // end click
    }); // end ready
```


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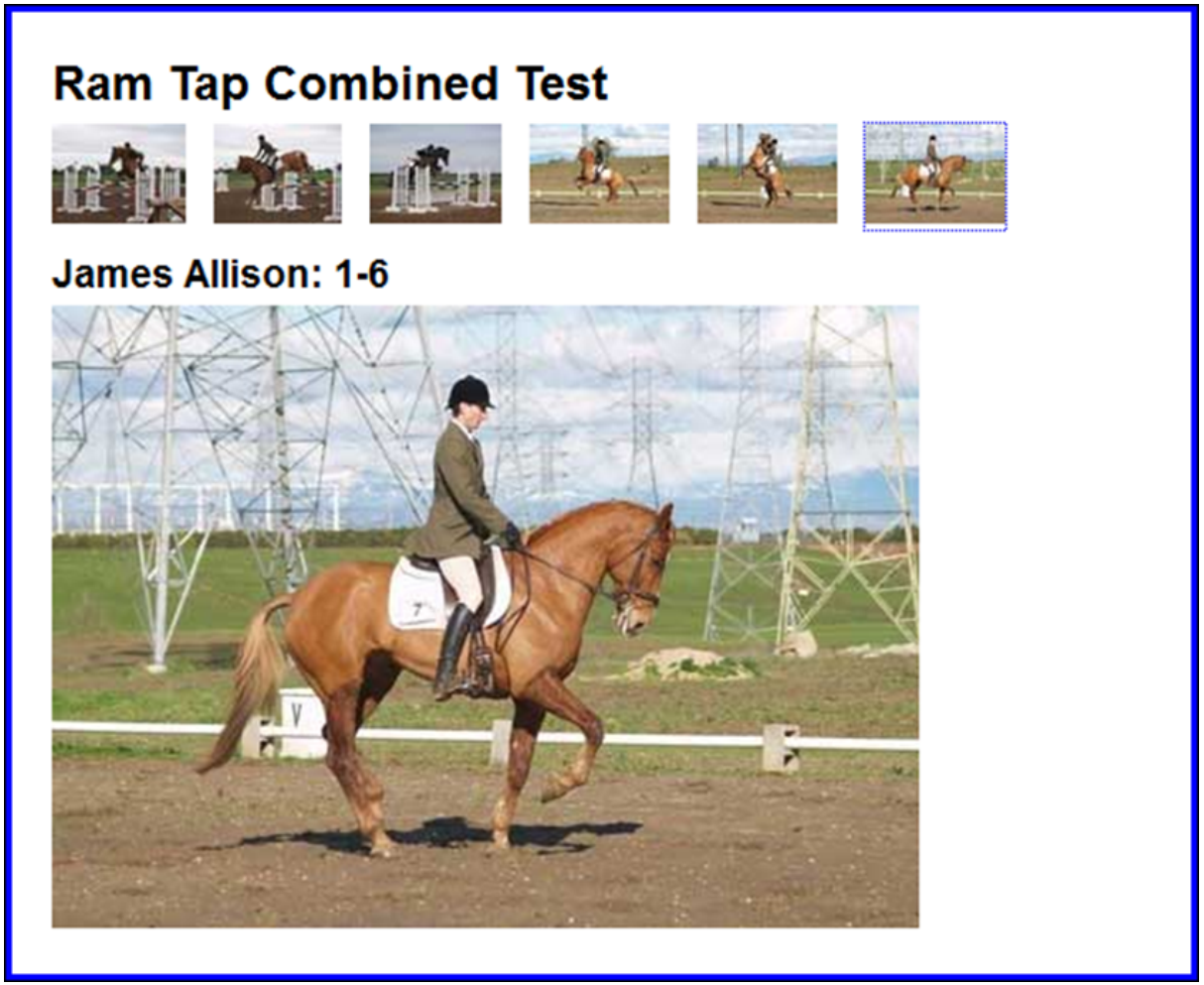
The user interface for the Image Swap application



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The script elements for the image swap

```
<script src="jquery-3.6.0.min.js"></script>
<script src="image_swaps.js"></script>
```


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The HTML for the images

```
<main>
    <h1>Ram Tap Combined Test</h1>
    <ul id="image_list">
        <li><a href="images/h1.jpg"
               title="James Allison: 1-1">
            <img src="thumbnails/t1.jpg" alt=""></a></li>
        <li><a href="images/h2.jpg"
               title="James Allison: 1-2">
            <img src="thumbnails/t2.jpg" alt=""></a></li>
        ...
        <li><a href="images/h6.jpg"
               title="James Allison: 1-6">
            <img src="thumbnails/t6.jpg" alt=""></a></li>
    </ul>
    <h2 id="caption">James Allison 1-1</h2>
    <p><img src="images/h1.jpg" alt="" id="image"></p>
</main>
```


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The CSS for the li elements

```
li {padding-right: 10px; display: inline; }
```


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A Slide Show application with fading out and in

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The script elements for the slide show

```
<script src="jquery-3.6.0.min.js"></script>
<script src="slide_show.js"></script>
```


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The HTML for the slide show

```
<section>
    <h1>Fishing Slide Show</h1>
    <h2 id="caption">Casting on the Upper Kings</h2>
    <img id="slide" src="images/casting1.jpg" alt="">
    <div id="slides">
        <img src="images/casting1.jpg"
             alt="Casting on the Upper Kings">
        <img src="images/casting2.jpg"
             alt="Casting on the Lower Kings">
        <img src="images/catchrelease.jpg"
             alt="Catch and Release on the Big Horn">
        <img src="images/fish.jpg"
             alt="Catching on the South Fork">
        <img src="images/lures.jpg"
             alt="The Lures for Catching">
    </div>
</section>
```


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The critical CSS for the slide show

```
#slides img { display: none; }
```


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Two websites for JavaScript and jQuery code

- Dynamic Drive
- The JavaScript Source

Two types of jQuery components that you can add to your website

- jQuery plugins
- jQuery UI widgets

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A jQuery plugin for a carousel called bxSlider

On June 6th, 2012, Vecta Corp. moved into its new 4 story, 35,000 square foot facilty. Below are a few pictures of the new facility.

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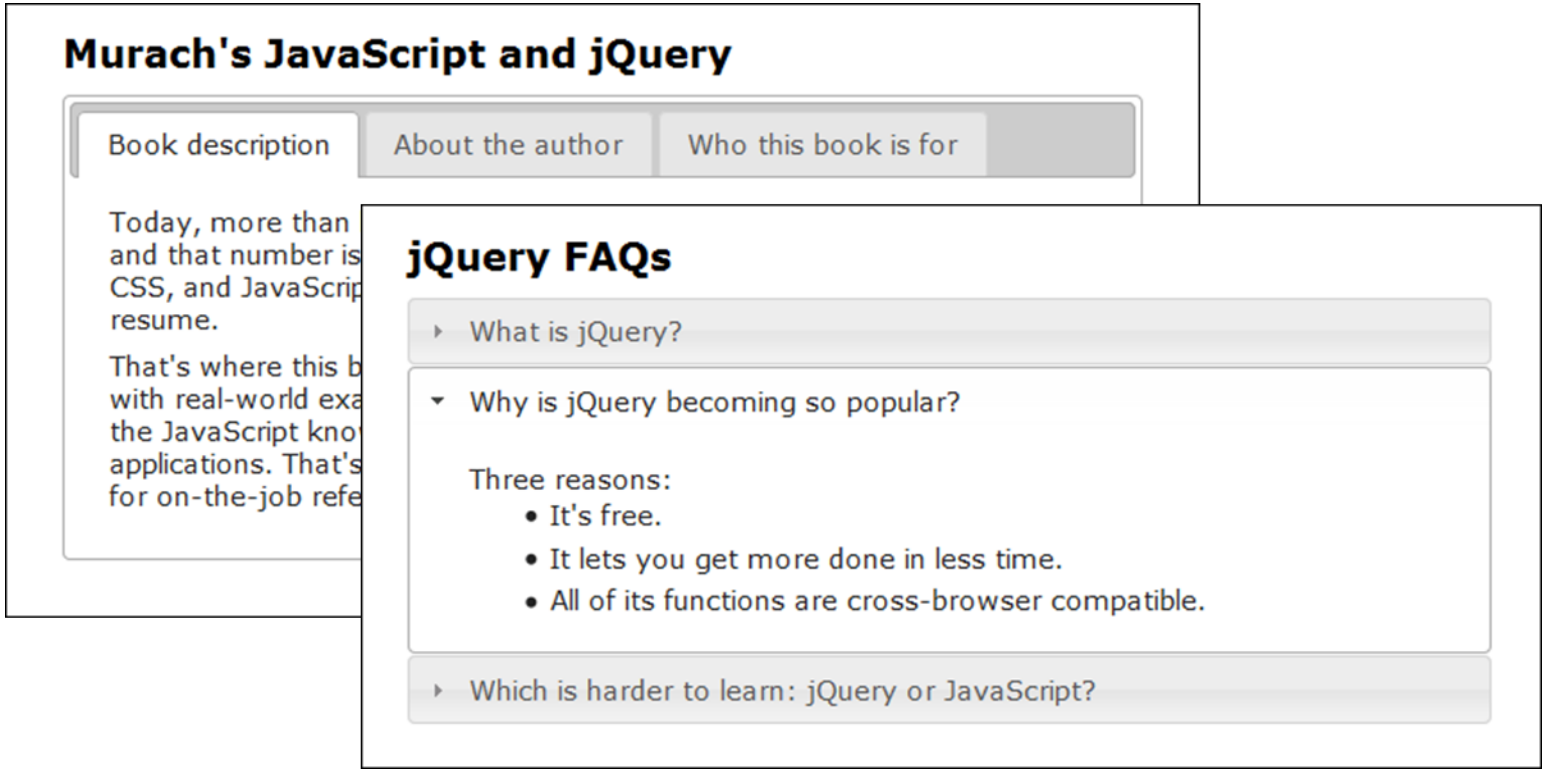
jQuery UI widgets for tabs and accordions



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Short 18-1 Do image rollovers using jQuery

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